



THE SUPERIOR UNIVERSITY LAHORE

Quiz - 1

Semester: 3rd

Section: C

Session: Fall 2025

Faculty of Computer Science and Information Technology

Subject: Data Structures and Algorithms

QCH:

Total Marks: 10

Name.....

Roll No.....

Instructions:

1. Cheating is not allowed, otherwise 0 marks will be rewarded.

Task #	CLO #	Domain and BT Level	Total Points
1	1	C6	05
2	1	C3	05

Task 1: Undo/Redo Operations

(5 Marks)

Stacks allow apps like text editors or drawing programs undo your last action or redo it if you change your mind.

- ✚ You type “Hi” in a text editor, then add “!”.
- ✚ Undo stack: [Type Hi, Add !].
- ✚ Press Undo: Pop “Add !”, text becomes “Hi”, and “Add !” goes to a redo stack.
- ✚ Press Redo: Move “Add !” back to the undo stack, text becomes “Hi!”.

Task 2: Call Center Simulation

(5 Marks)

A call center handles incoming customer calls on a first-come, first-served basis using a queue. Write a program that:

- ✚ Adds callers to the queue when they call (accept the caller's name as input).
- ✚ Removes a caller from the queue when their issue is resolved.
- ✚ Displays the next caller to be served without removing them from the queue.
- ✚ Displays the complete list of callers currently waiting in the queue.

Hint: Use enqueue for adding, dequeue for removing, and peek for the next caller.

~~ Best of Luck ~~